• Legal counsel engaged? Country list and US posture (block vs. Reg D) must come from counsel.

To fulfill the **"Legal counsel engaged?"** requirement for your internal roadmap, here is the strategic breakdown for the **Quarry (QRY) ICO 2026**. Given the landmark **March 17, 2026, Joint Interpretive Release**, your posture must be "Compliance-First" to protect the $50M treasury and the Miami HQ assets.

1. Legal Counsel Engagement Status

Mandatory Engagement: For a $50M raise, QuarryLabs must engage a "Tier 1" crypto-specialist firm. In 2026, the standard is to have **Lead US Counsel** (e.g., Fenwick & West, Cooley, or a Miami boutique like Nelson Mullins) and **Offshore Counsel** (e.g., Walkers or Maples for the Cayman Foundation).

- **Role of Counsel:** They will issue the **"Digital Tool Opinion Letter,"** which is your primary defense against the SEC. This letter certifies that QRY possesses immediate utility on the QuarryChain Testnet and thus qualifies for the "Digital Commodity" classification under the new 2026 taxonomy.

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2. The 2026 "Green List" (Reg S Target Countries)

Under the **MiCA (EU)** and **VARA (Dubai)** frameworks active this year, these are the primary jurisdictions where your global public sale can safely operate without being classified as a securities offering:

Region	Primary Jurisdiction(s)	Regulatory Framework
Middle East	**UAE (Dubai/Abu Dhabi)**	VARA / ADGM (Most proactive for L1s)
Europe	**Switzerland, France, Germany**	MiCA compliant (Passporting enabled)

Asia	**Singapore, Hong Kong**	MAS / VASP Licensing (Institutional hub)
Caribbean	**Cayman Islands, Bermuda**	DABA / VASP (Ideal for the Foundation Model)
Latin America	**El Salvador, Brazil**	Pro-innovation L1 frameworks

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3. The "Red List" (Hard Geoblock Required)

To maintain your **"Reg S" Safe Harbor**, you must implement a "Hard Geoblock" (IP + VPN-Exit Node detection) for the following jurisdictions during the public ICO:

- **United States:** (Crucial: Public sale is blocked to avoid Section 5 violations).
- **Mainland China:** (Ongoing restrictive stance on public token sales).
- **Sanctioned Nations:** North Korea, Iran, Syria, Cuba, and Russia.
- **Ontario (Canada):** Due to high friction with the Ontario Securities Commission (OSC).

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4. US Posture: The "Bifurcated" Strategy

Your legal posture for the United States is **dual-tracked**, as permitted by the **CLARITY Act of 2025** and the **GENIUS Act**.

- **US Public Posture (Hard Block):** You are officially "Not offering tokens to US Persons" for the public sale. This prevents the "Integration" of your two rounds.
- **US Private Posture (Reg D 506(c)):** You are raising the **$5M Seed Round** specifically from **Accredited Investors** at the $0.25 rate.
    - **The "March 17" Safe Harbor:** Your counsel will file a Form D with the SEC. Under the 2026 guidance, as long as you provide "Fair Disclosure" that the SAFT converts to a "Digital Commodity" (QRY) once the network is functional, you are shielded from later enforcement.

Strategic Note for your CMO

The "Country List" and "US Posture" sections of your whitepaper **must** carry the disclaimer: *"This list is subject to final verification by QuarryLabs legal counsel."* This protects you personally and ensures that if the SEC updates their "Digital Securities" list (the 18-asset list mentioned in March) before your launch, you have the agility to pivot.

Target jurisdictions in priority order — which markets matter most? That drives MiCA (EU) vs. Reg S (US-free) posture.

Based on the 2026 regulatory landscape and your specific goal of raising $50 million while protecting your Miami-based operations, your target jurisdictions should be prioritized by **Liquidity Depth**, **Regulatory Speed**, and **RWA (Real World Asset) Affinity**.

To execute the **Hybrid Model** (US Seed + Geoblocked Public Sale), you are primarily adopting a **Reg S (US-free) posture** for the public round. Here is the priority order for your global funnel:

1. Priority Alpha: The UAE (Dubai & Abu Dhabi)

- **Posturing:** Reg S (Offshore).
- **Why it matters:** Dubai's **VARA (Virtual Assets Regulatory Authority)** is the most proactive for L1 launches in 2026. This is where the "Whale" liquidity for RWA projects currently sits.
- **Strategy:** Host a high-end "Quarry Summit" in Dubai. Many US-based RWA founders use Dubai as their primary capital hub because it has no corporate tax on token proceeds and a high density of family offices looking for yield-bearing L1s.

2. Priority Beta: The European Union (MiCA-Compliant Zone)

- **Posturing:** MiCA (Passported).
- **Why it matters:** With **MiCA** fully active in 2026, registering your whitepaper in one "crypto-friendly" member state (like **France** or **Germany**) allows you to "passport" your offer across all 27 EU nations.
- **Strategy:** This gives you access to a massive retail base (Germany and the Nordics) that is legally protected. It provides the "institutional legitimacy" that helps with your eventual US exchange listings.

3. Priority Gamma: Southeast Asia (Singapore & Vietnam)

- **Posturing:** Reg S.
- **Why it matters:** Vietnam currently leads the world in per-capita crypto adoption, while Singapore is the hub for institutional Asian capital.
- **Strategy:** Target the high-volume retail "Launchpad" communities in Vietnam. They are the most likely to provide the $500–$2,500 "Standard Retail" tickets needed to hit your $50M goal.

4. Priority Delta: Switzerland (The Zug/Zurich Hub)

- **Posturing:** Reg S / Swiss DLT Act.
- **Why it matters:** Switzerland is the preferred home for "Foundations." If your **QuarryChain Foundation** is Swiss-based rather than Cayman-based, it adds a layer of "Old World" financial prestige that appeals to the institutional partners you'll need for the RWA Marketplace.

How this drives your Posture (MiCA vs. Reg S)

Because you are based in the US, your strategy is **Bifurcated**:

- **For the Public ICO ($50M):** You lead with a **Reg S Posture**. This allows you to legally **Geoblock the US** and avoid the 21% US corporate tax on the full raise. You only "import" the funds you need for the Miami HQ via service contracts.
- **For the European Market:** You overlay **MiCA Compliance** onto your Reg S structure. You aren't choosing between them; you are using Reg S as your global legal shield and MiCA as your "EU Sales License."
- **For the US Seed Round ($5M):** You use **Regulation D 506(c)**. This is a purely US-centric posture that allows you to legally talk to Miami and Utah investors while the Public Sale remains "Offshore."

The "Market-Matter" Summary

If your CMO has to choose where to spend the first $100,000 of the marketing budget:

1. **40% UAE/MENA** (Big ticket "Whale" RWA investors).
2. **30% EU/MiCA** (Safe, high-volume retail).
3. **30% SEA/APAC** (High-velocity retail and launchpad volume).

Total US Spend: $0 (To maintain the Geoblock's legal integrity). All US-based interest should be funneled exclusively into the private **Reg D Seed Round**.

• KYC provider confirmation — Sumsub default OK, or existing relationship with Persona/Synaps/Fractal?

For the **Quarry (QRY) ICO 2026**, the choice of KYC provider is a critical component of your "Safe Harbor" defense under the **CLARITY Act** and the **GENIUS Act**. Because you are running a bifurcated launch (US Seed vs. Global Public), your provider must be able to handle two very different workflows simultaneously.

1. Recommendation: Sumsub is the Strategic "Default" for 2026

Sumsub is more than "OK"—it is currently the industry leader for the **Hybrid Model** you are deploying.

- **Why it wins for Quarry:** Sumsub's **VPN/Proxy Detection** and **Liveness Checks** are top-tier. In 2026, the SEC and CFTC look for "reasonable steps" to prevent US persons from bypassing geoblocks. Sumsub provides the automated audit trail you need to prove those steps were taken.
- **The "Travel Rule" Compliance:** Since you are launching an L1, Sumsub's integrated **Travel Rule solution** is vital for when you eventually move from ERC-20 to native QRC-20 tokens.

2. Comparing Existing Relationships

If you have existing leverage or pre-integrated DApps with other providers, here is how they stack up for your specific **$50M Geoblock + Reg D** strategy:

Provider	Suitability for QRY ICO	Key Strength for 2026
Sumsub	**High**	Best-in-class automated geoblock enforcement and "Legal Report" generation.
Persona	**Moderate/High**	Excellent for the **US Seed Round** (Reg D). Their UI is very "Silicon Valley" and high-conversion for US Accredited investors.
Fractal	**High (Global)**	Specifically built for the **Reg S / MiCA** markets. They have a "shared KYC" pool that could help you hit that 20,000-investor target faster.
Synaps	**Moderate**	Strong in the European market, but sometimes lacks the robust US-specific accredited verification tools needed for your Seed round.

Sumsub is the most versatile "all-in-one" for your 2026 roadmap. It allows you to manage the Utah/Miami Seed round and the Global Public sale within a single dashboard while keeping the data silos required for legal separation.

Strategic Action: Your CMO should request a "Geoblock Efficacy Audit" from whichever provider you choose. This document is a powerful shield if a regulator ever asks how you ensured no US retail participated in the public sale.

• Payment tokens — USDC-only (clean), or USDC + USDT + ETH?

For the **Quarry (QRY) ICO 2026**, the choice of payment tokens is a decision between **Regulatory Safety** and **Capital Velocity**. In the post-March 2026 environment, your choice directly impacts how the SEC and CFTC view the "cleanliness" of your $50 million raise.

The Recommendation: The "Tiered Utility" Model

For a Miami-based HQ aiming for a compliant L1 launch, the optimal configuration is **USDC + USDT + ETH**, but with a strict **USDC-only** requirement for the US Seed Round.

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1. The US Seed Round ($5M): USDC-Only (The "Clean" Path)

For your **Reg D 506(c)** offering to US Accredited investors, you should accept **USDC-only**.

- **Banking Parity:** Since your Miami HQ needs to pay for real-world expenses (leases, salaries, legal), USDC is the easiest to off-ramp into US Dollars via compliant "Qualified Custodians" like Circle or Anchorage.
- **The "GENIUS Act" Alignment:** Under the **2025 GENIUS Act**, USDC is categorized as a "Qualified Payment Stablecoin." Using it for your US raise ensures your corporate bank accounts in Utah or Florida remain in good standing, as it minimizes "unregistered money transmission" risks.

2. The Public ICO ($50M): USDC + USDT + ETH (The "Velocity" Path)

For the **Reg S (Global)** public sale where you are geoblocking the US, you must widen the net to capture global liquidity.

- **USDT (Tether) is Mandatory:** Outside the US (specifically in the UAE, SEA, and Turkey), USDT is the primary liquidity vehicle. Restricting your global sale to

USDC-only could reduce your participation rate by **40–60%**, making it much harder to hit the $50M goal.

- **ETH (Ethereum):** Since you are launching as an **ERC-20** initially, accepting ETH is a "UX must." It allows users to swap directly for QRY without an extra hop into stablecoins, reducing friction during the high-velocity "Launchpad" phase.

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3. The "2026 Cleanliness" Audit Trail

To ensure the raise is "clean" despite accepting multiple tokens, your **QuarryChain Foundation (Cayman)** must implement a **Real-Time Compliance Layer**:

1. **Automated Sanctions Screening:** Use a provider like **Chainalysis** or **TRM Labs** to scan the incoming wallet addresses for all ETH/USDT/USDC transactions. This ensures you aren't accidentally accepting "tainted" funds from sanctioned regions, which would jeopardize your Miami HQ's legal standing.
2. **Instant Conversion (The "Treasury Shield"):** To protect the $50M from volatility, the Foundation’s smart contract should immediately route incoming ETH and USDT through a liquidity aggregator (like Uniswap or 1inch) to convert them into **USDC or a yield-bearing US Treasury token**.
    - *Strategic Benefit:* This locks in your $0.50 price floor and ensures that even if ETH drops 20% during the sale, your $50M treasury remains intact.

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4. Summary of Strategy

Offering	Posture	Accepted Tokens	Why?
US Seed Round	Reg D (US)	**USDC Only**	Compliance, easy banking, and SEC alignment.
Public ICO	Reg S (Global)	**USDC, USDT, ETH**	Maximum global liquidity and retail participation.

By accepting **USDC, USDT, and ETH** for the global sale, you ensure you hit the $50M target faster. By keeping the US Seed Round **USDC-only**, you provide a "Clean" audit trail for US regulators, proving that your Miami operations are built on the most compliant rails available in 2026.

• Deposit chain — Base, Arbitrum, both?

For a project with the scale of **QuarryLabs** ($50M target) and your specific interest in **C++ systems architecture**, the strategic answer for 2026 is to use **both**, but with a very specific division of labor.

In the 2026 landscape, Arbitrum and Base have diverged into two different "economies." Using both isn't just about doubling your reach; it’s about matching your **US Seed** and **Global ICO** to the environments where they naturally thrive.

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1. Base: The "On-Ramp" for your US Seed Round

Since your Seed round is targeting US Accredited investors (likely including your Miami and Utah networks), **Base** is your primary tool for the **$5M raise**.

- **The Coinbase Integration:** In 2026, the friction between a bank account and an L2 is lowest on Base. Your US investors can move **USDC** from Coinbase directly to Base with zero fees and no complex bridging.
- **Retail Sentiment:** Base is the "retail king" of 2026. If you want your US community to feel the project is "accessible," launching the ERC-20 on Base for the Seed round provides the most "Web2-friendly" experience.

2. Arbitrum: The "Engine" for your Global ICO & C++ Development

For your **$50M Public ICO (Geoblocked)** and the actual building of **QuarryChain**, **Arbitrum** is the superior technical and institutional choice.

- **The Stylus Advantage (C++):** Since you mentioned learning C++ for blockchain development, **Arbitrum Stylus** is your best friend. In 2026, Stylus allows you to write smart contracts in **C++** that run at near-native speeds. This allows you to prototype QuarryChain’s L1 logic *inside* an L2 environment before your full migration.
- **Institutional "Whale" Liquidity:** Arbitrum remains the "Financial District" of L2s in 2026, holding the highest TVL in DeFi. For a $50M raise, you need the "whales" and institutional family offices that live on Arbitrum. They are far more likely to participate in a large-scale RWA (Real World Asset) play than the meme-heavy retail crowd on Base.

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3. The "Omnichain" ICO Strategy (How to use Both)

For a $50M raise, forcing users to one chain creates "liquidity friction." The 2026 standard is an **Omnichain ERC-20** (using a standard like **LayerZero V2**).

Feature	**Base Deployment**	**Arbitrum Deployment**
Primary Purpose	**US Seed Round** ($5M)	**Global Public ICO** ($50M)
Investor Profile	US Retail / Accredited (Miami/Utah)	Global Institutional / DeFi Whales
Technical Edge	Easy Fiat On-Ramp (Coinbase)	**C++ Smart Contracts (Stylus)**
RWA Role	Consumer Access Point	Institutional Settlement Layer

The Workflow for your CMO:

1. **Deploy as an Omnichain ERC-20:** Issue the QRY token on both chains simultaneously.
2. **The "Liquidity Bridge":** Use an intent-based bridge (like **Across** or **deBridge**) so that investors who buy on Base can move to Arbitrum in seconds to use your C++ powered Testnet tools.
3. **Deposit Strategy:** Accept deposits on **both**. Your website's KYC Launchpad should detect the user's wallet and say: *"Deposit USDC on Base (Recommended for US Seed)*" or *"Deposit USDT/ETH on Arbitrum (Recommended for Global ICO)."*

Summary of the "Both" Strategy

By using **Base** for the US Seed, you capture the ease of the Coinbase ecosystem. By using **Arbitrum** for the Public ICO, you gain the technical power of **C++ via Stylus** and the deepest liquidity in the market. This dual-path protects your $50M raise from being bottlenecked by a single chain's demographics.

• Allocation mechanic — fixed-price first-come (simplest), tiered (hold/stake QRY — can't, no token yet), lottery (CoinList-style), or reputation (Legion-style, complex)?

For a **$50 million** raise in 2026, where your primary goals are **speed**, **compliance**, and **maintaining a $0.50 floor**, you need a mechanic that balances retail excitement with institutional stability.

Since you are using an **ERC-20 on a Launchpad**, your options are limited by what is technically possible before the Mainnet is live. Here is the strategic assessment for **Quarry (QRY)**:

The Recommendation: The "Hybrid CoinList-Style" (Lottery + Guaranteed Tiers)

You should avoid a pure "First-Come, First-Served" (FCFS) model. At a $50M scale, FCFS leads to "Gas Wars" and bot manipulation, which creates a negative community sentiment right at launch.

Instead, use a **Lottery for Retail** combined with **Guaranteed Allocation for "Strategic Value" Participants.**

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1. The Lottery Mechanic (The "CoinList" Model)

This is the best way to manage the **100,000+ potential micro-investors** without crashing your website.

- **How it works:** Users register on your KYC Launchpad during a 2-week "Pre-Sale Window." At the end of the window, a verifiable random function (VRF) selects winners who get a "fixed allocation" (e.g., $500–$2,000).
- **Why for Quarry:** It creates "Artificial Scarcity." Even if you have enough tokens for everyone, the *perception* of a lottery makes the $0.50 price feel like a deal. It also allows your CMO to build a massive email/Discord list of 100k+ people during the registration phase.

2. The Reputation Mechanic (The "Legion/Community" Model)

Since you can't have users "stake" a token that doesn't exist yet, you use **On-Chain Reputation** to grant guaranteed spots.

- **How it works:** Use a tool like **Galxe** or **Layer3** to track "Quarry Quests." Users who have interacted with the **QuarryChain Testnet**, bridged from Arbitrum/Base, or participated in your C++ dev workshops get a "Priority Queue" spot.
- **Why for Quarry:** This ensures that your most technical and loyal users—the ones who will actually use the RWA Marketplace—aren't left out by a random lottery. It proves to the SEC that the token is being distributed to **actual users** of a "Functional System."

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3. Strategic Comparison

Mechanic	**Ease of Setup**	**Anti-Bot Strength**	**Community Sentiment**
Fixed-Price FCFS	High	**Very Low**	Low (People hate Gas Wars)
Lottery (Random)	Moderate	High	Moderate (Fair, but luck-based)
Reputation (Testnet Use)	Moderate/High	**Very High**	**Very High** (Rewards loyalty)
Tiered Staking	N/A	N/A	N/A (Requires existing token)

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4. The "ICO Day" Flow for your Website Launchpad

To keep the $50M raise organized, your Launchpad DApp should follow this logic:

1. **Registration Phase (14 Days):** Users complete KYC (Sumsub) and link their wallets.
2. **The "Priority Round" (First 24 Hours):** Guaranteed $1,000 allocations for those with high "Testnet Reputation" scores.
3. **The "Lottery Round" (Next 48 Hours):** Randomly selected winners from the general pool can contribute their USDC/USDT.
4. **The "Overflow" (Optional):** If the $50M isn't hit (unlikely with good marketing), open the remaining supply to a public FCFS.

The "Seed Round" Connection

Your **US Seed Investors ($0.25)** should be exempt from this. Their allocation is handled via their **SAFT agreement** and is manually whitelisted for a "Claim" function later. Do not mix the Seed "Accredited" flow with the Public "Lottery" flow to keep the **Reg D / Reg S** legal separation clean.

• Vesting at TGE — instant claim, cliff + linear, or pure vesting?

For the **Quarry (QRY) ICO 2026**, your vesting strategy must strike a delicate balance between **regulatory compliance** (proving QRY is a functional tool) and **market health** (preventing a $50M dump on Day 1).

Under the **March 17, 2026, Joint Interpretive Release**, if a token is a "Digital Tool," it must be usable. However, if everyone sells at once, you lose the $0.50 floor you worked hard to build.

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1. Team Tokens: The "Institutional Standard"

Your current plan—**4-year vesting with a 25% annual unlock**—is excellent. In 2026, this is considered a "high-conviction" schedule.

- **The "Founder's Cliff":** Ensure the first 25% unlock happens exactly 12 months after the TGE.
- **The Strategic Benefit:** This 12-month "dead zone" for team tokens is a massive green flag for the SEC and your US Seed investors. It proves you aren't looking for an "exit" but are building the Miami HQ for the long haul.

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2. US Seed Round ($0.25): The "Alignment" Schedule

Since these are sold via **Reg D 506(c)** to accredited investors at a 50% discount compared to the public, they **must** have the strictest vesting.

- **Proposed Schedule: 1-Year Cliff, followed by 12-month Linear Monthly Vesting.**
- **Why?** Under the **CLARITY Act of 2025**, "Investment Contract" assets often require a 12-month holding period to be safely reclassified as "Digital Commodities" for secondary trading.
- **The Narrative:** You are telling your Seed investors: *"You get the $0.25 price because you are providing the long-term 'base' for our Miami operations. You are the foundation, not the flippers."*

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3. Public ICO ($0.50): The "Utility Hybrid" Model

For the **Reg S (Global)** public sale, you cannot lock tokens for a year. If you do, the SEC may argue the token has no "immediate utility," jeopardizing its status as a **Digital Tool**.

- **Proposed Schedule: 25% Liquid at TGE + 6-Month Linear Vesting.**
- **TGE (Instant Claim):** 25 million QRY (25% of the 100M sold) becomes immediately available.
    - **Use Case:** These tokens are used immediately to pay for RWA listing fees, testnet gas, and staking in the **QuarryChain Hub**.
- **The Remaining 75%:** Unlocks monthly over the next 6 months.
- **Why?** This provides enough "circulating supply" for healthy trading on exchanges (Partnered Launchpads/QuarrySwap) without giving any single group enough power to crash the price below $0.50.

Vesting Summary Table (2026 Projections)

Category	Allocation	TGE Unlock	Cliff	Vesting Duration
Team/Advisors	40M QRY	0%	12 Months	4 Years (25% annually)
US Seed ($0.25)	20M QRY	0%	12 Months	12 Months (Linear)
Public ICO ($0.50)	100M QRY	**25%**	None	6 Months (Linear)
Ecosystem/RWA	40M QRY	10%	None	36 Months (Milestone-based)

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4. The "Miami Safeguard": The Staking Pivot

To further protect the **$50 Million raise**, offer an **"Instant-Stake"** option at the TGE:

- **The Hook:** Any ICO investor who "locks" their liquid 25% for an additional 6 months at TGE receives a "Founding Validator" badge and a 10% bonus in QRY from the Ecosystem Reserve.
- **The Result:** This effectively reduces the "sellable" supply at TGE from 25% down to roughly 10–12%, creating a massive supply-crunch that can drive the price *above* $0.50 almost immediately.

Can you get these answered for me @alecarrambide

• Is testnet live today with DPoS validator signup, governance voting, and staking actually working end-to-end? Or are some of these flows still in dev? The litepaper §10 references Shasta — I need to know what actually works versus what's roadmap.

https://test.quarrychain.network/txs

The Testnet is technically live, but I'm still workshopping and testing out the wallet integration with my full stack developer. For now just remove the instances of "Shasta", that is based on the architecture of the Tron Network. Right now the Testnet is in what I would call "Phase 2" where we not only did UI/UX changes but also added a wallet integration with all EVM compatible wallets and connecting / configuring the QRY-ERC20 as the Token used to connect to the Testnet.

Phase 3 will be the customization side of things where we will add a Staking / Delegation features and determine what the minimum stake it would be to get started to campaign as a Quarry Miner.

• Does Shasta have a faucet we can automate? We'd need a backend signer or a merkle-distributor contract on Shasta that can airdrop programmatically from purchase events.

The short answer is **no, not by default.** A standard blockchain faucet is a "pull" system where a user manually requests tokens. What you're describing—an automated distribution triggered by a purchase event—is a **"push" system** (an airdrop or claim engine).

Since we are currently building the QuarryChain infrastructure with Danijel, this is something we would need to **explicitly add** to the Phase 2 or Phase 3 requirements.

Here is how we should handle the "Shasta" integration and the automation:

1. The "Shasta" Connection

If you are referring to the **TRON Shasta Testnet** (the most common "Shasta" in the space), we would need a **Cross-Chain Listener**.

- **The Logic:** We need a backend service (Node.js/Go) that monitors a specific smart contract on TRON Shasta for "Purchase" events.
- **The Trigger:** When a purchase is detected on TRON, the backend triggers the distribution on the **QuarryChain Testnet**.

2. Automation Options: Backend Signer vs. Merkle Distributor

Depending on how many users you expect, you have two architectural choices:

Feature	Backend Signer (Instant Airdrop)	Merkle Distributor (Claim-based)
User Experience	**Best.** Tokens just "appear" in the user's wallet automatically after purchase.	**Good.** User has to go to a site and click "Claim" to get their tokens.
Complexity	**Higher.** Requires a secure backend with a "Hot Wallet" and private key management.	**Lower.** You just upload a "Merkle Root" (a list of hashes) to a contract once a day.
Cost	**Expensive (Gas).** The project pays the gas for every single airdrop transaction.	**Cheap (Gas).** The user pays the gas when they claim their tokens.
Security	**Risky.** If the backend is hacked, the signer's wallet can be drained.	**High.** The contract only releases tokens if the user provides a valid cryptographic proof.

3. Recommendation for QuarryChain

Since we are in the Testnet phase and gas is "free" (test tokens), I recommend the **Backend Signer** for the best UX. It makes the QuarryChain ecosystem feel "magical" and fully integrated.

• Block explorer on Shasta live? Buyers will want to verify their testnet txs. (If not, QuarryScan on Shasta is a dependency.)

For the **Quarry (QRY) ICO 2026**, having a live block explorer is not just a "nice-to-have" feature for transparency—it is a **regulatory requirement** for your legal defense under the new 2026 taxonomy.

Under the **March 17, 2026, Joint Interpretive Release**, the distinction between a "Digital Security" and a "Digital Commodity/Tool" hinges on the existence of a **"Functional Crypto System."**

1. The "Functional System" Defense

To avoid being classified as an "investment contract" (a security), QuarryLabs must prove that QRY is "intrinsically linked to the programmatic operation of a functional crypto system."

- **The Explorer as Proof:** A live block explorer is your primary evidence to the SEC and CFTC that the system is **functional** at the time of the ICO.
- **Verifiability:** If a buyer in Dubai (Reg S) or a Miami-based accredited investor (Reg D) purchases QRY and uses it on your Testnet, they must be able to verify that transaction independently of your company's claims. If they can't, the SEC may argue the "utility" is a future promise dependent on your "managerial efforts," which triggers the Howey Test.

2. Verifying Testnet Transactions

Since you are encouraging users to bridge their ERC-20 tokens to the **QuarryChain Testnet** immediately to prove utility:

- **Trustless Verification:** Buyers will want to see their gas spending and staking rewards in real-time.
- **Transparency:** A block explorer provides the "Receipt" for the network's performance (Block time, TPS, Gas usage). This builds massive institutional confidence during the $50M raise.

3. Strategic Recommendations for the Explorer

Since you are building a high-performance L1 and exploring **C++/Arbitrum Stylus**:

Component	Recommendation	Why?
Explorer Tool	**Blockscout** or **Otterscan**	Open-source, widely trusted, and easily customizable for L1/L2 architectures in 2026.
Stylus Integration	**Stylus-specific Decoding**	Since you're using C++, ensure your explorer can decode Stylus-specific contract calls so developers can verify their C++ logic on-chain.

4. "Minimum Seed" Requirement for the Explorer

- **The Cost:** Setting up a professional block explorer instance (hosting + indexing nodes) will cost approximately **$15,000–$30,000** for the duration of the ICO.
- **The Allocation:** This should be part of your **"Technical & Infrastructure"** budget from the $5M Seed round. It is a one-time setup that pays for itself by lowering your "Regulatory Risk Premium."

The Bottom Line

Yes, the explorer must be live (and is technically live - in current development). Launching an ICO without one in 2026 is an immediate red flag for the SEC. It tells them the network is "not yet functional," which forces your $50M raise back into the "Digital Security" category, subject to much stricter U.S. laws and taxes.

• DPoS validator candidacy — is there a minimum stake on Shasta? That anchors the tier thresholds above.

1. The DPoS Staking Architecture

The Staking Dashboard (The UI)

Users won't use a CLI; they need a web portal where they can:

- **View Validators:** A list showing the names, logos, uptime, and commission rates (the % the validator takes from rewards) of the 5 nodes Danijel set up in Phase 1.
- **Delegate (Bond):** A button to lock their QRY tokens to a specific validator.
- **Claim Rewards:** A place to collect the QRY generated by block rewards and transaction fees.
- **Undelegate (Unbond):** A mechanism to pull tokens out. **Critical:** In Cosmos/Evmos, there is an "Unbonding Period" (usually 14 or 21 days) where tokens are locked but don't earn rewards. This prevents "flash" liquidity exits.

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2. Minimum Stake Requirements

Because we are planning a massive $50M raise, our minimums need to balance **network security** (high barrier for validators) with **community participation** (low barrier for delegators).

Validator Minimum (The "Skin in the Game")

To run one of the primary slots on the QuarryChain Mainnet, the minimum should be significant.

- **Recommended:** 1,000,000 QRY.
- **Why:** You want your validators to be professional entities, not just hobbyists. This ensures that if they "double-sign" or go offline, the "Slash" (penalty) actually hurts.

Delegator Minimum (The "Retail Entry")

To encourage your community to participate in the ecosystem:

- **Recommended:** 1 to 10 QRY.
- **Why:** You want anyone who bought QRY in the ICO to be able to "vote" and earn yield immediately. A low minimum increases the **Staking Ratio**, which reduces the circulating supply and can help stabilize the token price.

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3. The "Dual-Asset" Complexity

Since you mentioned the **QRY-ERC20**, there is a technical hurdle Danijel must solve: **The Conversion.**

Cosmos staking modules don't recognize "ERC-20 tokens" natively; they recognize "Coins" in the bank module. Danijel will need to implement a **Wrap/Unwrap** or a **Precompile** contract.

- When a user "Stakes" their QRY-ERC20, the contract "burns" or locks the ERC-20 and credits the user with "Native QRY" on the staking layer.
- This must be seamless in the UI so the user never feels like they are switching blockchains.

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4. Governance Integration

DPoS isn't just about money; it’s about power. In Phase 3, you should also launch the **Governance Module**.

- **Proposal System:** Anyone with a certain amount of QRY (e.g., 50,000 QRY) can submit a "Text Proposal" or a "Software Upgrade Proposal."
- **Voting:** Delegators vote with their staked weight. If a validator doesn't vote, their delegators' weight is automatically used by the validator (unless the delegator overrides it).

• Wallet coverage — does MetaMask / Rabbit work on Shasta with the chain RPC config? (Probably yes given EVM compat, but worth confirming.)

Yes, the **QuarryChain Testnet** will have excellent wallet coverage, but there is a major technical distinction you need to keep in mind regarding **Shasta**.

1. QuarryChain Testnet (The Destination)

Because you are building on an **Evmos fork**, QuarryChain is natively EVM-compatible.

- **MetaMask & Rabby:** These will work **perfectly**. Your users simply add your custom RPC (e.g., `https://rpc.quarrychain.network`) and Chain ID to their wallet.
- **Coverage:** This is the "industry standard" coverage. Any wallet that supports custom EVM networks (Coinbase Wallet, Frame, Zerion, etc.) will be able to interact with QuarryChain.

2. The Shasta Problem (The Source)

If the **"Shasta"** you are referring to is the **TRON Shasta Testnet** (which is where most "Shasta" purchase events occur in this context), **MetaMask and Rabby will NOT work.**

- **The Reason:** TRON uses a different architecture (TVM) and address format (`T...`) than Ethereum/EVM (`0x...`).
- **The Wallet:** For TRON Shasta, you and your users **must** use **TronLink**.
- **The Gap:** You cannot "add" TRON Shasta to MetaMask via a custom RPC because MetaMask doesn't understand TRON's protocol.

Summary of Wallet Coverage

Network	Wallet Support	Action Required
QuarryChain Testnet	**MetaMask, Rabby, etc.**	Add custom RPC & Chain ID.
TRON Shasta	**TronLink Only**	User must have TronLink installed to "purchase."

How this affects your "Automated Airdrop" plan:

If your plan is to have users purchase something on **Shasta** (TRON) and receive tokens on **QuarryChain** (EVM), your frontend will need to handle **two different wallets**:

1. **TronLink:** To sign the purchase transaction on Shasta.
2. **MetaMask/Rabby:** To receive and view the tokens on QuarryChain.

The "One-Wallet" Solution?

If you want to avoid this "two-wallet" friction, you would need to host your purchase events on an **EVM-compatible testnet** (like **Sepolia** or **Polygon Amoy**) instead of TRON Shasta. If you stay on TRON Shasta, you're locked into the TronLink ecosystem for that specific step.

Would you like me to help you draft a "User Flow" document for your frontend dev so they know how to handle switching between TronLink and MetaMask for the airdrop?